

MACHINE LEARNING PROJECT

AI-Powered Banking Fraud Detection

Transaction Risk Analysis & Predictive Classification
System

Context & Problem Statement

| The Banking Fraud Threat

Financial institutions lose billions annually to credit card fraud. Traditional rule-based security mechanisms are static and fail to adapt to evolving, complex fraud techniques.

| The Machine Learning Answer

By leveraging deep historical transaction signatures, machine learning models analyze latent variables in real-time, detecting highly anomalous behavioral patterns dynamically.

The Imbalance Challenge

0.17%

Fraudulent Transactions

A Needle in a Haystack

Out of 284,807 transaction entries, only 492 are labeled as fraud (99.83% genuine). This extreme class imbalance renders standard accuracy metrics misleading and requires target model optimizations aimed strictly at catching risk.

Dataset Class Proportions



The visual distribution highlights why standard Accuracy fails. A naive classifier predicting all transactions as genuine would achieve 99.83% accuracy, but catch exactly 0% of fraud.

Exploratory Risk Insights

Behavioral Patterns

Data analysis indicates distinct trends between fraudulent and normal activity. Transaction amounts show highly unusual spikes and outlier distributions for anomalous cases, while transaction velocities vary heavily with cyclic day-night patterns.



Project Methodology

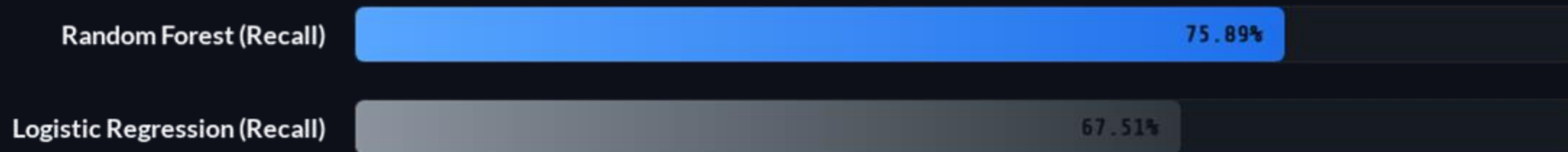
 **Train-Test Partitioning:** The historical logs were split into 80% train and 20% validation sets (227k vs. 56k entries) to avoid over-optimistic validation.

 **Metric Prioritization:** Instead of generic accuracy, the system optimizes for Recall. We maximize caught fraud while keeping false alarms within bounds.

 **Mathematical Evaluation:**

$$\text{Recall} = \frac{TP}{TP + FN}$$

Hyperparameter Optimization



Through 3-fold cross-validated GridSearchCV, the Random Forest model (configured with a maximum depth of 10 and 50 estimators) outperformed Logistic Regression on key Recall parameters by 8.38%.

Model Classification Report

Class Target	Precision	Recall	F1-Score	Support
Genuine (Class 0)	1.00	1.00	1.00	56,864
Fraud (Class 1)	0.95	0.76	0.84	98
Weighted Avg	1.00	1.00	1.00	56,962

Confusion Matrix breakdown

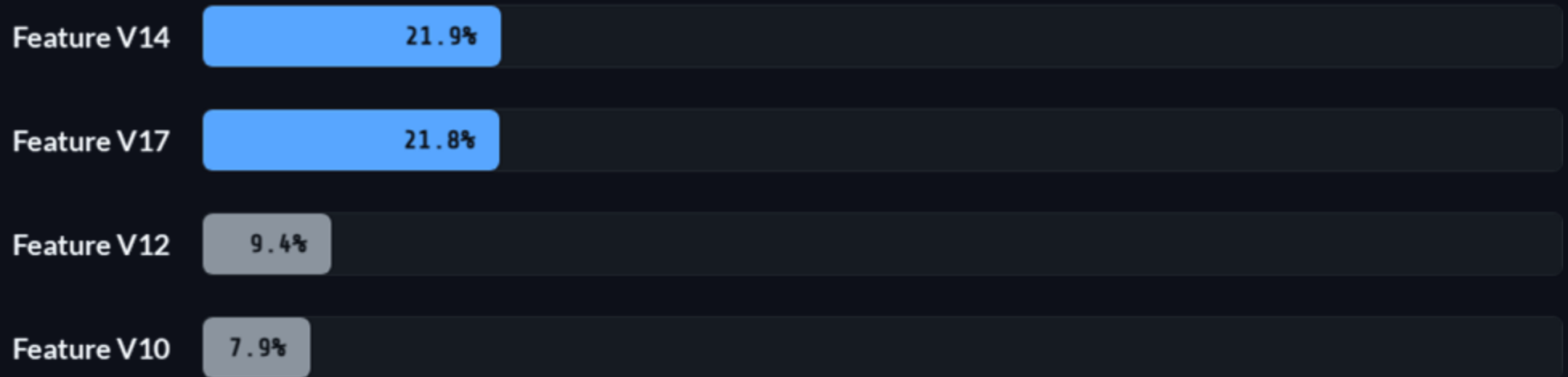
| Detection Success

The final tuned Random Forest correctly identified 74 out of 98 fraud cases (True Positives), and flagged only 4 false alarms (False Positives) out of 56k+ clean cases, proving high precision.

| The Threat of Leaks

However, 24 fraudulent transactions bypassed the classifier (False Negatives). While 76% Recall is state-of-the-art on raw logs, further behavior profiling is recommended to plug the remaining leak.

Top Predictive Features



The top risk indicators, V14 and V17 (latent behavioral representations calculated via PCA), represent over 43% of the total predictive power of the model. Prioritizing these variables speeds up real-time risk scores.

Deployment & Scale

Real-Time API

The trained Random Forest was serialized using joblib into a lightweight pipeline artifact (fraud_detection_model.pkl).

This model is ready for real-time scoring in transaction authorization loops, providing an instantaneous risk probability API for modern web or mobile payment applications.

THANK YOU